# An explicit two-sided residue decomposition for sparse-paving Kazhdan-Lusztig comparisons

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*Editorial note.* Complete original author text, analytic arguments, low-rank certificate tables and embedded bibliography retained; one primary per task. Original mathematics unchanged, no proof repair or independent correctness certification.

*Allowed standard prerequisites (editorial).*

- Matroid Kazhdan-Lusztig polynomials, real-rootedness and residue comparisons. Native polynomial definitions, circuit-hyperplane correction formula, all essential analytic comparisons and exact low-rank certificate tables retained in full.. No essential endpoint or kernel argument is removed as a prerequisite; all original author arguments remain available.

## 1 Introduction

Elias, Proudfoot, and Wakefield [6] introduced the Kazhdan–Lusztig polynomial $P_M(x)$ of a matroid $M$ and conjectured that its coefficients are nonnegative and form a log-concave sequence. They proved nonnegativity for representable matroids by interpreting the coefficients as dimensions of intersection cohomology groups. Braden, Huh, Matherne, Proudfoot, and Wang [1] proved the nonnegativity conjecture for all matroids using singular Hodge theory. Gedeon, Proudfoot, and Young [12] proposed the stronger conjecture that the zeros are all real and negative. Cheng and Liu [4] constructed representable matroids with nonunimodal Kazhdan–Lusztig polynomials, so neither log-concavity nor real-rootedness holds for arbitrary matroids.

A matroid of rank $d$ is *sparse paving* if every $d$-element subset of its ground set is either a basis or a circuit-hyperplane. This class contains all uniform matroids. Mayhew, Newman, Welsh, and Whittle [15] conjectured that asymptotically almost all matroids are sparse paving.

Let $P_{m,d}(x)$ denote the Kazhdan–Lusztig polynomial of the uniform matroid of rank $d$ on $m+d$ elements. Proudfoot, Wakefield, and Young [17] computed $P_{1,d}(x)$ and interpreted its coefficients using polygon dissections. Gedeon, Proudfoot, and Young computed the equivariant polynomials of uniform matroids [11] and proved real-rootedness for corank one [12, Theorem 3.3]. Gao, Lu, Xie, Yang, and Zhang [10] obtained the positive formula for the coefficients that we use below and proved real-rootedness for coranks $2 \leq m \leq 15$ and all ranks using multiplier sequences and symbolic computations. Xie and Zhang [19] proved log-concavity for all uniform matroids. Gao, Li, Xie, Yang, and Zhang [9] recovered this log-concavity from induced log-concavity of the equivariant Kazhdan–Lusztig polynomials.

Proudfoot, Xu, and Young [18] introduced the $Z$-polynomial of a matroid. Xie and Zhang [20] proved its real-rootedness for all sparse paving matroids and strict interlacing for uniform matroids of fixed positive corank in consecutive ranks.

For sparse paving matroids, Lee, Nasr, and Radcliffe [14] gave a coefficient formula in terms of skew Young tableaux and proved nonnegativity. Their formula shows that $P_M(x)$ depends only on the rank, the corank, and the number of circuit-hyperplanes. Ferroni and Vecchi [8] obtained this dependence by relaxing circuit-hyperplanes and computationally verified real-rootedness for sparse paving matroids with at most 30 elements. Ferroni, Nasr, and Vecchi [7] obtained formulas for the Kazhdan–Lusztig polynomials of paving matroids by relaxing stressed hyperplanes. Xie and Zhang [21] proved unimodality for loopless sparse paving matroids and determined the modes. Our main result is as follows.

**Theorem 1.1.** *Every nonconstant Kazhdan–Lusztig polynomial of a sparse paving matroid has only simple negative zeros.*

By Newton's inequalities, Theorem 1.1 implies log-concavity of the coefficients. Our proof uses a formula for relaxing circuit-hyperplanes. Relaxing a circuit-hyperplane means declaring it to be a basis. We use the convention that $P_M(x) = 0$ when $M$ has a loop [8, Remark 2.8]. In particular, $P_{1,0}(x) = 0$. Write $\Delta_d(x) = P_{1,d}(x) - P_{1,d-1}(x)$. Ferroni and Vecchi [8, Corollary 3.8 and Theorem 5.2] showed that a sparse paving matroid $M$ of rank $d \geq 1$ and corank $m$, with $\lambda$ circuit-hyperplanes, satisfies

$$P_M(x) = P_{m,d}(x) - \lambda \Delta_d(x).$$

Set

$$\bar{\lambda}_{m,d} = \frac{\binom{m+d}{d}}{\max(m,d)+1}.$$

The bound on the number of circuit-hyperplanes in [8, Lemma 5.4] gives $0 \leq \lambda \leq \bar{\lambda}_{m,d}$. For $d \geq 5$, write $n = \lfloor (d-1)/2 \rfloor$. The following theorem applies to every real parameter satisfying this upper bound.

**Theorem 1.2.** *Let $d \geq 5$ and $m \geq 2$ be integers, and let $\lambda$ be real with $\lambda \leq \bar{\lambda}_{m,d}$. Then $P_{m,d}(x) - \lambda \Delta_d(x)$ has degree $n$ and only simple negative zeros, and $P_{2,d-2}(x)$ strictly interlaces this polynomial.*

For a fixed rank, Theorem 1.2 gives a common strict interlacer for all coranks $m \geq 2$ and all real parameters $\lambda \leq \bar{\lambda}_{m,d}$. Section 2 establishes the strict interlacing that we use in the proof and reduces the problem to inequalities at the zeros of $P_{2,d-2}(x)$. The sign of the correction term shows that it suffices to consider the largest allowed value of $\lambda$. We state the required endpoint inequality in Proposition 2.6.

Section 3 proves the endpoint inequality for $d \geq 17$. A partial fraction expansion and two residue comparisons reduce the proof to three sign conditions. We first prove positivity of the coefficient at the negative pole closest to zero. We combine the terms at the positive poles into an integral representation with a positive decreasing weight. The quotient of this sum by $P_{2,d-2}(x)$ has positive residues at the zeros of $P_{2,d-2}(x)$. An integral lower bound controls the derivative of the weight and supplies the comparison needed in a recurrence in the corank. A direct estimate handles the remaining small coranks.

Section 4 proves the endpoint inequality for $5 \leq d \leq 16$. We take remainders modulo $P_{2,d-2}(x)$ and expand in powers of $m-2$. The coefficients in this expansion are polynomials in $x$. We verify their signs by exact computation on rational intervals containing the zeros of $P_{2,d-2}(x)$. These signs give the required inequalities for every $m \geq 2$. We conclude with the proofs of the main theorems, treating ranks at most four and coranks at most one separately.

## 2 Reduction to the endpoint inequality

Throughout the rest of the paper, $d \geq 5$ and $m \geq 2$ are integers unless we state otherwise. We reduce Theorem 1.2 to a sign inequality at the zeros of $P_{2,d-2}(x)$ when $\lambda = \bar{\lambda}_{m,d}$. We first use residues and coefficient identities to prove that $P_{1,d-2}(x)$ and $P_{2,d-2}(x)$ strictly interlace. This interlacing shows that decreasing $\lambda$ preserves the required signs at the zeros of $P_{2,d-2}(x)$. We then verify the degree and leading coefficient needed to count all zeros and state the endpoint inequality. Sections 3 and 4 prove that inequality.

Let $p(x)$ and $q(x)$ be nonzero real polynomials of degrees $j$ and $k$, respectively, with only simple real zeros. Write their zeros as $\rho_1 < \cdots < \rho_j$ and $\sigma_1 < \cdots < \sigma_k$, respectively. If $k = j+1$ and $j \geq 1$, we write $p(x) \prec q(x)$ when

$$\sigma_1 < \rho_1 < \sigma_2 < \cdots < \rho_j < \sigma_{j+1}.$$

If $k = j \geq 1$, we write $p(x) \prec q(x)$ when

$$\rho_1 < \sigma_1 < \cdots < \rho_j < \sigma_j.$$

We say that $p(x)$ and $q(x)$ strictly interlace if $p(x) \prec q(x)$ or $q(x) \prec p(x)$. For $j = 0$ and $k = 1$, we take $p(x) \prec q(x)$ by convention. We use the following characterization of strict interlacing in terms of residues. See also [13].

**Lemma 2.1.** *Let $p(x)$ and $q(x)$ be real polynomials with positive leading coefficients. Suppose that $p(x)$ has degree $j \geq 1$ and simple real zeros $\rho_1 < \cdots < \rho_j$.*

*(i) If $\deg q(x) = j$, then $p(x) \prec q(x)$ if and only if*

$$\frac{q(\rho_i)}{p'(\rho_i)} < 0 \qquad \text{for every } i.$$

*(ii) If* $\deg q(x) = j$*, then* $q(x) \prec p(x)$ *if and only if*

$$\frac{q(\rho_i)}{p'(\rho_i)} > 0 \qquad \text{for every } i.$$

*(iii) If* $\deg q(x) = j-1$*, then* $q(x) \prec p(x)$ *if and only if*

$$\frac{q(\rho_i)}{p'(\rho_i)} > 0 \qquad \text{for every } i.$$

*Proof.* The residue of $q(x)/p(x)$ at $\rho_i$ is

$$\frac{q(\rho_i)}{p'(\rho_i)}.$$

Since $p'(\rho_i)$ has sign $(-1)^{j-i}$, the stated interlacing orders imply the corresponding residue signs. Conversely, each prescribed set of residue signs gives a zero of $q(x)$ between each pair of consecutive zeros of $p(x)$. When $\deg q(x) = j$, we use the positive leading coefficient to obtain $q(x) > 0$ for sufficiently large positive $x$ and $(-1)^j q(x) > 0$ for sufficiently large negative $x$. Negative residues give $q(\rho_j) < 0$, so we obtain one further zero to the right of $\rho_j$. Positive residues give $(-1)^{j-1}q(\rho_1) > 0$, so we obtain one to the left of $\rho_1$. When $\deg q(x) = j-1$, the intervals between consecutive zeros of $p(x)$ already account for all zeros of $q(x)$. By counting degrees in each case, we obtain simple zeros and the asserted interlacing. □

Gedeon, Proudfoot, and Young [12, Theorem 3.3] proved real-rootedness of $P_{1,d-2}(x)$. We use the coefficient identities below to construct diagonal operators that give simple negative zeros for $P_{1,d-2}(x)$ and $P_{2,d-2}(x)$ and preserve strict interlacing.

We have $P_{m,d}(0) = 1$. By [10, Theorem 1.3], the coefficient of $x^k$ in $P_{m,d}(x)$ is

$$\frac{\binom{m+d}{k}}{d-k} \sum_{h=0}^{m-1} \binom{d-k+h}{h+k+1} \binom{k-1+h}{h}, \quad 1 \le k \le n. \tag{2.1}$$

In particular, $\deg P_{m,d}(x) = n$ and $\deg P_{1,d-2}(x) = \deg P_{2,d-2}(x) = n-1$. Specializing (2.1) at coranks one and two gives

$$P_{2,d-2}(x) = dP_{1,d-2}(x) - \Delta_d(x)/x. \tag{2.2}$$

To prove that the diagonal operators below preserve simple real zeros, we use the following consequence of [5, Theorem 3.1].

**Lemma 2.2.** *Let* $\gamma_0, \ldots, \gamma_{n-1} > 0$*, and define the linear map* $T$ *by*

$$T\left(\sum_{k=0}^{n-1} p_k x^k\right) = \sum_{k=0}^{n-1} \gamma_k p_k x^k.$$

*If the Jensen polynomial*

$$\sum_{k=0}^{n-1} \binom{n-1}{k} \gamma_k x^k$$

*has only negative real zeros, then* $T$ *sends nonzero real polynomials of degree at most* $n-1$ *with simple real zeros to polynomials of the same degree with simple real zeros. Moreover, if* $p(x)$ *and* $q(x)$ *have degree at most* $n-1$ *and* $p(x) \prec q(x)$*, then* $T(p(x)) \prec T(q(x))$*.*

*Proof.* We apply Theorem 3.1 of [5] to see that $T$ preserves real-rootedness. Since every $\gamma_k > 0$, $T$ preserves degree and is a linear homeomorphism on each $\mathbb{R}[x]_{\le j}$, $j \le n-1$. A polynomial of degree $j$ with simple real zeros lies in the interior of the set of real-rooted polynomials in $\mathbb{R}[x]_{\le j}$, whereas one with a multiple zero lies on its boundary. If $p(x)$ has degree $j$ and simple real zeros, we choose an open neighborhood $U$ of $p(x)$ in $\mathbb{R}[x]_{\le j}$ consisting of real-rooted polynomials. Then $T(U)$ is an open neighborhood of $T(p(x))$ that contains only real-rooted polynomials. Since $T$ preserves degree, $T(p(x))$ also has simple real zeros.

Let $p(x)$ and $q(x)$ strictly interlace. By Obreschkoff's theorem [2, Theorem 8], every nonzero real linear combination of $p(x)$ and $q(x)$ has simple real zeros. We use linearity, injectivity, and the preceding paragraph to obtain the same property for $T(p(x))$ and $T(q(x))$. We apply the theorem once more to obtain strict interlacing.

If $p(x)$ and $q(x)$ have different degrees, we use degree preservation to determine the interlacing order. Suppose they have the same degree $j \geq 1$ and $p(x) \prec q(x)$. By Vieta's formulas, $T$ multiplies each sum of zeros by $\gamma_{j-1}/\gamma_j > 0$. The sum of the zeros of $p(x)$ is smaller than that of $q(x)$. Since $T(p(x))$ and $T(q(x))$ strictly interlace, we use this comparison to conclude that $T(p(x)) \prec T(q(x))$.

When the input polynomial has only negative zeros, $T$ preserves coefficient signs and real-rootedness, so the image also has only negative zeros. □

We now describe the finite multiplier sequences that we will use. If $p(x) = \sum_{k=0}^{n-1} p_k x^k$ has degree $n-1$, simple negative zeros, and positive leading coefficient, then $p_k > 0$ and the sequence $p_k/\binom{n-1}{k}$, $0 \leq k \leq n-1$, has Jensen polynomial $p(x)$. We also use reciprocals of factorials as multipliers. For each integer $s \geq 0$, set

$$\mathscr{L}_s(x) = \sum_{k=0}^{n-1} \binom{n-1}{k} \frac{x^k}{(k+s)!}.$$

We evaluate the generalized Laguerre polynomial of degree $n-1$ and parameter $s$ at $-x$ to obtain a positive multiple of $\mathscr{L}_s(x)$. Thus $\mathscr{L}_s(x)$ has simple negative zeros [16, Sections 18.2(vi) and 18.5(iii)]. The sequences $1/(k+s)!$ and $1/(n+s-1-k)!$, indexed by $0 \leq k \leq n-1$, have Jensen polynomials $\mathscr{L}_s(x)$ and $x^{n-1}\mathscr{L}_s(1/x)$, respectively. Lemma 2.2 therefore applies to the corresponding diagonal operators. Compositions of these operators preserve the conclusions of the lemma.

We write $\vartheta_x = x\,d/dx$ for the Euler operator. Rational functions of $\vartheta_x$ act coefficientwise by substituting $k$ for $\vartheta_x$ on the monomial $x^k$. The denominators in these operators do not vanish at the relevant coefficient indices. We apply an operator before evaluating the resulting polynomial. We also write $(u)_j = u(u+1)\cdots(u+j-1)$, with $(u)_0 = 1$.

The preservation of simple real zeros by the following two operators is a special case of Brändén [3, Corollary 3.5]. We include a direct proof of the strict interlacing orders.

**Lemma 2.3.** *Let $f(x)$ be a nonconstant real polynomial with simple negative zeros and positive leading coefficient. If $\alpha > 0$ and $\beta > \deg f(x)$, then $(\alpha + \vartheta_x)f(x)$ and $(\beta - \vartheta_x)f(x)$ have the same degree as $f(x)$, have only simple negative zeros, and satisfy*

$$(\beta - \vartheta_x)f(x) \prec f(x) \prec (\alpha + \vartheta_x)f(x).$$

*Proof.* We use the zeros and positive leading coefficient of $f(x)$ to see that all its coefficients are positive. The two operators multiply its coefficient of $x^k$ by $\alpha+k$ and $\beta-k$, respectively, so both resulting polynomials have positive coefficients and the same degree as $f(x)$. At a zero $r < 0$ of $f(x)$, their quotients by $f(x)$ have residues $r < 0$ and $-r > 0$, respectively. We apply Lemma 2.1 to obtain the asserted strict interlacing and simple real zeros. Since both polynomials have positive coefficients, we conclude that all their zeros are negative. □

For $\alpha > 0$ and $\beta > n-1$, both operators also satisfy the hypotheses of Lemma 2.2. Their Jensen polynomials of degree $n-1 \geq 1$ are

$$(1+x)^{n-2}\big(\alpha + (\alpha+n-1)x\big), \qquad (1+x)^{n-2}\big(\beta + (\beta-n+1)x\big).$$

Thus we may compose these operators with the preceding diagonal operators.

Set

$$a_d = \frac{\sqrt{d^2+8} - d}{2}, \qquad b_d = d + a_d.$$

Then $0 < a_d < 2/d < 1$, $a_d b_d = 2$, and $u(d-u) + 2 = (u + a_d)(b_d - u)$. The zeros $-a_d$ and $b_d$ of this quadratic lie outside the interval $[0, n-1]$.

For each integer $j \geq 0$, let $U_j(z)$ denote the Chebyshev polynomial of the second kind. It satisfies

$$U_j(\cos\theta) = \frac{\sin((j+1)\theta)}{\sin\theta}, \qquad 0 < \theta < \pi,$$

by [16, Eq. (18.5.2)]. Set

$$f_d(x) = \sum_{k=0}^{n-1} \binom{d-k-3}{k} x^k.$$

The coefficient expansion

$$U_{d-3}(z) = \sum_{k=0}^{n-1} (-1)^k \binom{d-k-3}{k} (2z)^{d-3-2k}$$

gives

$$f_d(-1/(4z^2)) = (2z)^{-(d-3)}U_{d-3}(z). \tag{2.3}$$

The coefficient formula (2.1) gives

$$P_{1,d-2}(x) = (d-1)!\sum_{k=0}^{n-1} \frac{\binom{d-k-3}{k}}{(k+1)!(d-1-k)!}x^k, \tag{2.4}$$

and

$$P_{2,d-2}(x) = d!\sum_{k=0}^{n-1} \frac{(k+a_d)(b_d-k)\binom{d-k-3}{k}}{(k+2)!(d-k)!}x^k. \tag{2.5}$$

By comparing the coefficients in (2.4) and (2.5), we find that, for $0 \le k \le n-1$, the ratio of the coefficient of $x^k$ in $P_{2,d-2}(x)$ to that in $P_{1,d-2}(x)$ is

$$\frac{d(k(d-k)+2)}{(k+2)(d-k)}. \tag{2.6}$$

**Lemma 2.4.** *For every integer $d \ge 5$, the polynomials $f_d(x)$, $P_{1,d-2}(x)$, and $P_{2,d-2}(x)$ have $n-1$ simple negative zeros, and $P_{1,d-2}(x) \prec P_{2,d-2}(x)$. In particular, each zero $r$ of $P_{2,d-2}(x)$ satisfies*

$$\frac{\Delta_d(r)}{rP'_{2,d-2}(r)} = d\frac{P_{1,d-2}(r)}{P'_{2,d-2}(r)} > 0. \tag{2.7}$$

*Proof.* Using (2.3), we obtain all $n-1$ zeros of $f_d(x)$ from the positive zeros of $U_{d-3}(z)$. The resulting zeros of $f_d(x)$ are distinct and negative. By (2.5), we obtain $(a_d+\vartheta_x)^{-1}P_{2,d-2}(x)$ by multiplying $f_d(x)$ by $d!$, dividing its coefficient of $x^k$ by $(k+2)!(d-k)!$, and applying $b_d-\vartheta_x$. We obtain $(b_d-\vartheta_x)^{-1}P_{2,d-2}(x)$ in the same way, with $a_d+\vartheta_x$ as the final operator. Lemma 2.2 applies because $a_d > 0$ and $b_d > n-1$. Both polynomials therefore have degree $n-1$, simple negative zeros, and positive leading coefficients. We apply Lemma 2.3 to these two polynomials to obtain $n-1$ simple negative zeros for $P_{2,d-2}(x)$ and

$$(a_d+\vartheta_x)^{-1}P_{2,d-2}(x) \prec P_{2,d-2}(x) \prec (b_d-\vartheta_x)^{-1}P_{2,d-2}(x).$$

Their quotients by $P_{2,d-2}(x)$ have positive and negative residues, respectively.

By (2.6), we have

$$P_{1,d-2}(x) = \frac{1}{d}P_{2,d-2}(x) + \frac{2}{d(a_d+b_d)}\left(\frac{b_d-1}{a_d+\vartheta_x} - \frac{a_d+1}{b_d-\vartheta_x}\right)P_{2,d-2}(x).$$

Since $b_d-1>0$ and $a_d+1>0$, the displayed identity gives

$$\frac{P_{1,d-2}(r)}{P'_{2,d-2}(r)} > 0$$

at every zero $r$ of $P_{2,d-2}(x)$. Both $P_{1,d-2}(x)$ and $P_{2,d-2}(x)$ have degree $n-1$ and positive leading coefficients. Lemma 2.1(ii) therefore gives $P_{1,d-2}(x) \prec P_{2,d-2}(x)$. This also gives $n-1$ simple negative zeros for $P_{1,d-2}(x)$. Finally, we obtain the equality in (2.7) from (2.2). □

To determine the degree, (2.2) and (2.6) give the ratio of the coefficient of $x^{k+1}$ in $\Delta_d(x)$ to that of $x^k$ in $P_{1,d-2}(x)$ as

$$\frac{2d(d-k-1)}{(k+2)(d-k)}, \qquad 0 \le k \le n-1. \tag{2.8}$$

In particular, $\Delta_d(x)/x$ has positive coefficients. At the upper endpoint $\lambda = \bar{\lambda}_{m,d}$, set

$$F_{m,d}(x) = P_{m,d}(x) - \bar{\lambda}_{m,d}\Delta_d(x).$$

**Proposition 2.5.** *For integers $d \ge 5$ and $m \ge 2$, the polynomial $F_{m,d}(x)$ has degree $n$ and positive leading coefficient.*

*Proof.* Both $P_{m,d}(x)$ and $\Delta_d(x)$ have degree at most $n$. We prove the proposition by showing that

$$[x^n]F_{m,d}(x) = [x^n]P_{m,d}(x) - \bar{\lambda}_{m,d}[x^n]\Delta_d(x) > 0.$$

We use the coefficient estimates in the proofs of [14, Lemmas 21, 22, and 25], taking $i = n \geq 2$ in their notation. For $m = 2$, we use

$$\bar{\lambda}_{2,d} = \frac{d+2}{2}$$

as in the proof of Lemma 22.

For $n = 2$ and $m \geq d$, we use

$$\bar{\lambda}_{m,d} = \frac{1}{m+1}\binom{m+d}{d}$$

as in the first case of the proof of Lemma 25.

For $n \geq 3$ and $m \geq 3$, and for $n = 2$ and $3 \leq m < d$, we use

$$\bar{\lambda}_{m,d} \leq \frac{2}{m+d+2}\binom{m+d}{d}$$

as in the proof of Lemma 21 and the second case of the proof of Lemma 25, respectively.

We combine these estimates with (2.4) and (2.8) to obtain the desired positivity, except possibly when $(m, d) = (3, 5)$. In that case, (2.1) and (2.8) give

$$[x^2]F_{3,5}(x) = 56 - \frac{28}{3} \cdot 5 = \frac{28}{3} > 0.$$

□

The preceding facts reduce Theorem 1.2 to the following inequality.

**Proposition 2.6.** *For integers $d \geq 5$ and $m \geq 2$, every zero $r$ of $P_{2,d-2}(x)$ satisfies*

$$\frac{F_{m,d}(r)}{rP'_{2,d-2}(r)} > 0. \tag{2.9}$$

Equation (2.7) shows that (2.9) gives the required residue signs for every $\lambda \leq \bar{\lambda}_{m,d}$. Proposition 2.5 and the positive coefficients of $\Delta_d(x)/x$ provide the degree and leading coefficient needed to apply Lemma 2.1(ii) with $xP_{2,d-2}(x)$ as the denominator.

# 3 The endpoint inequality for $d \geq 17$

In this section, we prove Proposition 2.6 for $d \geq 17$. In Subsection 3.1, we use a partial fraction expansion and two residue comparisons to reduce the proof to the three conditions of Proposition 3.2. In Subsection 3.2, we prove Proposition 3.5, which verifies condition (i). In Subsection 3.3, Lemma 3.6 gives a quantitative lower bound for $S_{m,d}(z)$ that verifies condition (ii) and supplies the estimate for $S_{m,d}(1)$ needed in the last subsection. In Subsection 3.4, we combine this estimate with Lemma 3.9 and a direct estimate for small coranks to prove Proposition 3.8, which verifies condition (iii).

Unless we state otherwise, $d \geq 5$ and $m \geq 2$ are integers.

## 3.1 A criterion for the endpoint inequality

The following expansion separates the contributions from the negative and positive poles. Throughout this section, we use the coefficients that we construct in the proof of the lemma.

**Lemma 3.1.** *For integers $d \geq 5$ and $m \geq 2$, there are real coefficients $c^{\mathrm{L}}_{m,d}(p)$ and $c^{\mathrm{R}}_{m,d}(p)$ such that*

$$\begin{aligned}\frac{F_{m,d}(x) - P_{2,d-2}(x)}{x} = \Bigg(-4 &+ \sum_{p \in \{a_d, 3, \ldots, m+1\}} \frac{c^{\mathrm{L}}_{m,d}(p)}{p + \vartheta_x} \\ &+ \sum_{p \in \{a_d, 1, \ldots, m-1\}} \frac{c^{\mathrm{R}}_{m,d}(p)}{d + p - \vartheta_x} - \frac{6(d-3)(d-1)}{d(3d-7)(d-3-\vartheta_x)}\Bigg) P_{2,d-2}(x).\end{aligned} \tag{3.1}$$

*Moreover, $c^{\mathrm{L}}_{m,d}(i) > 0$ for $3 \leq i \leq m$.*

To control the alternating coefficients at the positive integer poles, we combine the terms in (3.1) with denominators $b_d - \vartheta_x$ and $d + j - \vartheta_x$ into an integral. Set

$$g(t) = -\frac{1}{md\binom{m+d}{d}} \sum_{p \in \{a_d, 1, \dots, m-1\}} c^{\mathrm{R}}_{m,d}(p) e^{-(d+1+2p)t/2}.$$

Their sum in (3.1) is

$$-md\binom{m+d}{d} \int_0^\infty e^{-(d-1)t/2} g(t) P_{2,d-2}(e^t x)\, dt.$$

When $g(t)$ is positive and nonincreasing, Lemma 3.4 and the negative factor give positive residues for the quotient of this sum by $P_{2,d-2}(x)$. Since $g(t) \to 0$, it suffices to prove $-g'(t) > 0$. Define

$$E_{m,d} = -g'(0),$$
$$S_{m,d}(z) = \frac{1}{2md\binom{m+d}{d}} \sum_{j=1}^{m-1} c^{\mathrm{R}}_{m,d}(j)(d+1+2j)(j-a_d) z^{j-1}.$$

Differentiating $g(t)$ gives

$$-g'(t) = e^{-(a_d+b_d+1)t/2} \left( E_{m,d} + \int_{e^{-t}}^1 z^{-a_d} S_{m,d}(z)\, dz \right). \tag{3.2}$$

The following criterion reduces (2.9) to three sign conditions.

**Proposition 3.2.** *Let $d \geq 5$ and $m \geq 2$ be integers. Suppose that*

*(i)* $c^{\mathrm{L}}_{m,d}(a_d) > 0$,

*(ii)* $S_{m,d}(z) \geq 0$ *for every* $z \in [0,1]$,

*(iii)* $E_{m,d} > 0$.

*Then every zero $r$ of $P_{2,d-2}(x)$ satisfies*

$$\frac{F_{m,d}(r)}{r P'_{2,d-2}(r)} > 0.$$

We now prove the expansion and the two residue comparisons needed for the criterion.

*Proof of Lemma 3.1.* For integers $d \geq 5$ and $m \geq 1$, define the rational function

$$R_{m,d}(u) = \frac{(m+d)!(d-2-u)}{(d-1)!} \sum_{h=0}^{m-1} \frac{1}{h!(u+h+1)(u+h+2)(d-u+h)_{m-h}}. \tag{3.3}$$

We compare the coefficients in (2.1) and (2.4) with indices $k+1$ and $k$, respectively, to obtain

$$P_{m,d}(x) = 1 + x R_{m,d}(\vartheta_x) P_{1,d-2}(x).$$

Set

$$\tau_d(u) = \frac{(u+2)(d-u)}{d(u(d-u)+2)}, \qquad \nu_d(u) = \frac{(d-u)(d-3-2u)(d-4-2u)((u+1)(d-u-1)+2)}{(u+1)(u+3)(d-3-u)(u(d-u)+2)}.$$

We use the coefficient ratio (2.6) and compare consecutive coefficients in (2.5) to obtain

$$P_{1,d-2}(x) = \tau_d(\vartheta_x) P_{2,d-2}(x),$$
$$\frac{P_{2,d-2}(x) - 1}{x} = \nu_d(\vartheta_x) P_{2,d-2}(x).$$

The second identity includes the highest coefficient index $k = n-1$, since one of the factors $d-3-2k$ and $d-4-2k$ vanishes. We combine these identities with (2.2) and the representation of $P_{m,d}(x)$ above to obtain

$$\frac{F_{m,d}(x) - P_{2,d-2}(x)}{x} = \left(\tau_d(\vartheta_x)(R_{m,d}(\vartheta_x) - d\bar{\lambda}_{m,d}) + \bar{\lambda}_{m,d} - \nu_d(\vartheta_x)\right) P_{2,d-2}(x). \tag{3.4}$$

We expand

$$\tau_d(u)(R_{m,d}(u) - d\bar{\lambda}_{m,d}) + \bar{\lambda}_{m,d} - \nu_d(u)$$

into partial fractions. We let $c^{\mathrm{L}}_{m,d}(p)$ denote the coefficient of $(u+p)^{-1}$ for $p \in \{a_d, 3, \dots, m+1\}$, and let $c^{\mathrm{R}}_{m,d}(p)$ denote the coefficient of $(d+p-u)^{-1}$ for $p \in \{a_d, 1, \dots, m-1\}$. In particular, $p = a_d$ gives the nonintegral poles $-a_d$ and $b_d = d + a_d$.

We first compute the residues at the negative integer poles. For $1 \le i \le m$, only the summands with $h = i-1$ and, when $i \ge 2$, $h = i-2$ in (3.3) contribute at $u = -i$. Their residues sum to

$$\frac{(m+d)!(d+i-2)(d+i-1)(d+2i-3)!}{(d-1)!(i-1)!(d+m+i-1)!} > 0.$$

At $u = -m-1$, only $h = m-1$ contributes, with residue

$$-md\binom{m+d}{d}\frac{d+m-1}{d+2m}.$$

At $u = -1$, both $\tau_d(u)R_{m,d}(u)$ and $\nu_d(u)$ have residue $-(d+1)$, so their difference has no pole there. The factor $\tau_d(u)$ cancels the pole at $-2$. For $i \ge 3$, we have

$$\tau_d(-i) = \frac{(i-2)(d+i)}{d(i(d+i)-2)} > 0.$$

The only additional contribution at these integer poles comes from $-\nu_d(u)$ at $u = -3$, with positive residue

$$\frac{(d+1)(d+2)(d+3)^2}{d(3d+7)}.$$

At these integer poles, $c^{\mathrm{L}}_{m,d}(i) > 0$ for $3 \le i \le m$. Among the coefficients at negative integer poles, only $c^{\mathrm{L}}_{m,d}(m+1)$ can be negative.

For $0 \le j \le m-1$, only the summands with $0 \le h \le j$ in (3.3) contribute at $u = d+j$. We expand $(1-t)^j$ and integrate each term against $t^{d+j}(1-t)$ to express the coefficient of $(d+j-u)^{-1}$ as

$$\frac{(-1)^{j+1}(m+d)!(j+2)}{(d-1)!j!(m-1-j)!}\int_0^1 t^{d+j}(1-t)^{j+1}\,dt.$$

The factor $\tau_d(u)$ cancels the pole at $d$. For the remaining positive integer poles, we evaluate the beta integral and multiply by $\tau_d(d+j)$ to obtain

$$c^{\mathrm{R}}_{m,d}(j) = (-1)^{j+1}\frac{(m+d)!j(j+1)(j+2)(d+j+2)(d+j)!}{d!(m-1-j)!(d+2j+2)!(j(d+j)-2)}, \qquad 1 \le j \le m-1. \tag{3.5}$$

These coefficients alternate in sign.

We expand $\tau_d(u)$ into partial fractions as

$$\tau_d(u) = \frac{1}{d} + \frac{2(b_d-1)}{d(a_d+b_d)(u+a_d)} - \frac{2(a_d+1)}{d(a_d+b_d)(b_d-u)}.$$

For the pole at $-a_d$, set

$$\rho_d = \frac{db_d(a_d+b_d-4)(a_d+b_d-1)}{2(3(a_d+b_d)-11)}.$$

This constant is positive. We compute the coefficient of $(u+a_d)^{-1}$ in $\nu_d(u)$ as

$$\frac{2(b_d-1)\rho_d}{d(a_d+b_d)}.$$

We substitute this residue into (3.4) to obtain

$$c^{\mathrm{L}}_{m,d}(a_d) = \frac{2(b_d-1)}{d(a_d+b_d)}\big(R_{m,d}(-a_d) - d\bar{\lambda}_{m,d} - \rho_d\big). \tag{3.6}$$

For the pole at $b_d$, set

$$\phi_d = \frac{(a_d+1)(a_d+b_d+1)}{d(a_d+b_d)}, \qquad \eta_d = \frac{(a_d+b_d+1)^2(a_d+b_d+3)(a_d+b_d+4)}{b_d(a_d+b_d)(3(a_d+b_d)+11)}.$$

These constants are positive. We compute the coefficient of $(b_d - u)^{-1}$ in $\nu_d(u)$ as

$$-\frac{2\eta_d}{(a_d + b_d + 1)(b_d + 1)},$$

and we substitute this value into (3.4) to obtain

$$c^{\mathrm{R}}_{m,d}(a_d) = -\frac{2}{a_d + b_d + 1}\left(\phi_d\big(R_{m,d}(b_d) - d\bar{\lambda}_{m,d}\big) - \frac{\eta_d}{b_d + 1}\right). \tag{3.7}$$

The term with denominator $d - 3 - u$ comes from $-\nu_d(u)$ and has the coefficient in (3.1).

The rational functions corresponding to the operators in (3.1) and (3.4) have only simple poles. They have the same residues at all possible poles, and both tend to $-4$ as $u \to \infty$. They therefore coincide. To obtain (3.1), we substitute $\vartheta_x$ for $u$ and apply the operators to $P_{2,d-2}(x)$. □

To control the possibly negative coefficient at $-m-1$, we normalize the polynomials $(p+\vartheta_x)^{-1}P_{2,d-2}(x)$ to have the same leading coefficient as $P_{2,d-2}(x)$, then compare the residues of their quotients by $P_{2,d-2}(x)$. For $p > 0$, the required normalization factor is $n + p - 1$.

**Lemma 3.3.** *Let $d \geq 5$ be an integer. For each zero $r$ of $P_{2,d-2}(x)$, the sequence indexed by $p = a_d, 3, 4, \ldots$,*

$$(n + p - 1)\frac{(p + \vartheta_x)^{-1}P_{2,d-2}(x)\big|_{x=r}}{P'_{2,d-2}(r)},$$

*is positive, strictly decreasing, and converges to zero.*

*Proof.* For $n = 2$, we write $P_{2,d-2}(x) = 1 - x/r$ and obtain $-r/p$ for the displayed expression. Since $r < 0$, we obtain the assertion and may henceforth assume $n \geq 3$.

For successive parameters $(p, q) = (a_d, 3)$ or $(i, i + 1)$ with $i \geq 3$, we have

$$\frac{n + p - 1}{u + p} - \frac{n + q - 1}{u + q} = (q - p)\frac{n - 1 - u}{(u + p)(u + q)}.$$

We apply Lemma 2.1 to reduce strict decrease to

$$(n - 1 - \vartheta_x)((p + \vartheta_x)(q + \vartheta_x))^{-1}P_{2,d-2}(x) \prec P_{2,d-2}(x).$$

At each zero of $\mathscr{L}'_{q-1}(x)$, the identity $(p+\vartheta_x)\mathscr{L}_{q-1}(x) = p\mathscr{L}_{q-1}(x)+x\mathscr{L}'_{q-1}(x)$ shows that $(p+\vartheta_x)\mathscr{L}_{q-1}(x)$ has the same sign as $\mathscr{L}_{q-1}(x)$. We apply Rolle's theorem and compare leading coefficients to obtain

$$\mathscr{L}'_{q-1}(x) \prec (p + \vartheta_x)\mathscr{L}_{q-1}(x).$$

Define the linear operator $T$ on $\mathbb{R}[x]_{\leq n-1}$ to be diagonal in the monomial basis and to satisfy

$$T\big((p + \vartheta_x)\mathscr{L}_{q-1}(x)\big) = P_{2,d-2}(x).$$

Both polynomials have positive coefficients in every degree from 0 to $n - 1$, so this identity uniquely determines $T$. The matrix of $T$ in the monomial basis has positive diagonal entries. By (2.5), the map that multiplies the coefficient of $x^k$ by $1/(d - k)!$ and the operator $b_d - \vartheta_x$ send $d!f_d(x)$ to the Jensen polynomial of $T$ when $(p, q) = (a_d, 3)$. For $(p, q) = (i, i + 1)$, we additionally apply

$$(a_d + \vartheta_x)\prod_{j=3}^{i-1}(j + \vartheta_x).$$

These operations give simple negative zeros, so Lemma 2.2 applies to $T$. Using $(k + 1)\binom{n-1}{k+1} = (n - 1 - k)\binom{n-1}{k}$, we compare coefficients to obtain

$$T\big(\mathscr{L}'_{q-1}(x)\big) = (n - 1 - \vartheta_x)((p + \vartheta_x)(q + \vartheta_x))^{-1}P_{2,d-2}(x).$$

We apply $T$ to obtain strict interlacing. The polynomial on the left has degree $n - 2$ and positive leading coefficient, which proves the strict decrease of the sequence. Finally, for $0 \leq k \leq n - 1$, we have

$$\frac{n + p - 1}{k + p} \to 1 \qquad \text{as the integer } p \to \infty.$$

Thus the terms tend to

$$\frac{P_{2,d-2}(r)}{P'_{2,d-2}(r)} = 0.$$

From strict decrease to zero, we also obtain their positivity. □

For the positive poles, we use the following comparison for a positive nonincreasing weight.

**Lemma 3.4.** *Let $d \geq 5$ be an integer and $c > (d-2)/2$. Let $g \in C^1([0,\infty))$ be positive and nonincreasing. Then every zero $r$ of $P_{2,d-2}(x)$ satisfies*

$$\frac{1}{P'_{2,d-2}(r)} \int_0^\infty e^{-ct} g(t) P_{2,d-2}(e^t r)\, dt < 0.$$

*In particular, taking $g(t) = 1$ gives the corresponding inequality for $(c - \vartheta_x)^{-1} P_{2,d-2}(x)$.*

*Proof.* We first prove the assertion with $f_d(x)$ in place of $P_{2,d-2}(x)$. Since $c > \deg f_d(x)$ and $g$ is positive and bounded, we integrate each term to obtain a polynomial of the same degree with positive coefficients. Fix $z \in (0,1)$ with $U_{d-3}(z) = 0$ and put $r = -1/(4z^2)$. We substitute $u = ze^{-t/2}$ in (2.3) to obtain the weight

$$w(u) = u^{2c-d+2} g(2\log(z/u)), \qquad 0 < u \leq z.$$

Since $2c - d + 2 > 0$ and $0 < g(t) \leq g(0)$, we have $\lim_{u \to 0^+} w(u) = 0$. Differentiating $w(u)$, we obtain

$$w'(u) = u^{2c-d+1}\big((2c-d+2)g(2\log(z/u)) - 2g'(2\log(z/u))\big) > 0.$$

Let $T_{d-2}(z)$ be the Chebyshev polynomial of the first kind, so $T_{d-2}(\cos\theta) = \cos((d-2)\theta)$ for $0 < \theta < \pi$ [16, Eq. (18.5.1)]. At the zero we fixed, $T_{d-2}(z) = \pm 1$. Using $T'_{d-2}(z) = (d-2)U_{d-3}(z)$ [16, Eq. (18.9.21)], we integrate by parts to obtain

$$T_{d-2}(z) \int_0^z w(u) U_{d-3}(u)\, du = \frac{1}{d-2} \int_0^z w'(u)(1 - T_{d-2}(z) T_{d-2}(u))\, du > 0.$$

The integrand is nonnegative because $|T_{d-2}(u)| \leq 1$. Since $w'(u) > 0$ and $T_{d-2}(u)$ is nonconstant, the integrand is positive on a subinterval. The improper integrals converge since $\int_0^z w'(u)\, du = w(z) < \infty$. Reversing the substitution, we find that the integral with $f_d(x)$ in place of $P_{2,d-2}(x)$ has the sign of $T_{d-2}(z)$ at $r$. At the zero we fixed, we have

$$U'_{d-3}(z) = -\frac{(d-2)T_{d-2}(z)}{1-z^2}$$

by [16, Eq. (18.9.22)]. Differentiating (2.3), we see that $f'_d(r)$ has the opposite sign. Lemma 2.1 then gives

$$f_d(x) \prec \int_0^\infty e^{-ct} g(t) f_d(e^t x)\, dt.$$

We apply the diagonal operator in (2.5). It commutes with the substitution $x \mapsto e^t x$ and with integration, and it preserves strict interlacing by Lemma 2.2. We then apply Lemma 2.1(i) to obtain the required signs for $P_{2,d-2}(x)$. When $g(t) = 1$, we integrate each monomial to identify the integral with $(c - \vartheta_x)^{-1} P_{2,d-2}(x)$. □

*Proof of Proposition 3.2.* Using $b_d = d + a_d$, we integrate the monomials in $S_{m,d}(z)$ to obtain (3.2). Conditions (ii) and (iii), together with (3.2), give $-g'(t) > 0$ for every $t \geq 0$. Since $g(t) \to 0$, the weight is positive. We apply Lemma 3.4 with $c = (d-1)/2$ to the integral involving $g(t)$, and with $c = d - 3$ and $g(t) = 1$ to the term at the pole $d - 3$. Both values of $c$ exceed $(d-2)/2$. The sum of the terms corresponding to the positive poles in (3.1), including the term at $d - 3$, therefore has a negative leading coefficient and, after division by $P_{2,d-2}(x)$, positive residues.

Since $F_{m,d}(0) = P_{2,d-2}(0) = 1$, the left-hand side of (3.1) is a polynomial. Proposition 2.5 shows that it has degree $n - 1$ and positive leading coefficient. Since the sum corresponding to the positive poles has negative leading coefficient, we compare leading coefficients in (3.1) to obtain

$$\frac{c^{\mathrm{L}}_{m,d}(a_d)}{n + a_d - 1} + \sum_{i=3}^{m+1} \frac{c^{\mathrm{L}}_{m,d}(i)}{n + i - 1} > 4.$$

Fix a zero $r$ of $P_{2,d-2}(x)$. The sequence in Lemma 3.3 attains its positive minimum over $p \in \{a_d, 3, \ldots, m+1\}$ at $p = m + 1$. Condition (i) and Lemma 3.1 show that all other coefficients $c^{\mathrm{L}}_{m,d}(p)$ are positive. We replace the values for $p < m+1$ by this minimum and leave the term with $p = m+1$ unchanged. This gives a lower bound equal to the minimum times the sum in the preceding inequality. Hence the sum corresponding to the negative poles has positive residues after division by $P_{2,d-2}(x)$. Since $-4P_{2,d-2}(r) = 0$, we combine the two contributions to obtain (2.9). □

## 3.2 Positivity of the coefficient at $-a_d$

To verify condition (i) of Proposition 3.2, we construct a lower bound for $R_{m,d}(-a_d) - d\bar{\lambda}_{m,d}$. A recurrence for $R_{m,d}(u)$ shows that this lower bound increases with $m$, so it suffices to show that its value at $m = 2$ exceeds $\rho_d$.

**Proposition 3.5.** *For integers $d \geq 9$ and $m \geq 2$, we have $c^{\mathrm{L}}_{m,d}(a_d) > 0$.*

*Proof.* For every integer $d \geq 5$, we split off the last summand in (3.3) to obtain the recurrence

$$(m+d-u-1)R_{m,d}(u) = (m+d)R_{m-1,d}(u) + md\binom{m+d}{d}\frac{d-2-u}{(m+u)(m+u+1)}, \qquad m \geq 2. \tag{3.8}$$

For $m \geq 2$, we set

$$G_m = R_{m,d}(-a_d) - d\binom{m+d}{d}\left(\frac{1}{m+1} + \frac{2}{m(m-1)}\left(\frac{1}{d+1} - \frac{1}{m+1}\right)\right).$$

Since

$$0 < \frac{2}{m(m-1)} \leq 1,$$

the parenthesized expression is a convex combination of $1/(m+1)$ and $1/(d+1)$. Hence

$$G_m \leq R_{m,d}(-a_d) - d\bar{\lambda}_{m,d}.$$

By (3.6), it suffices to prove

$$G_m \geq G_2 > \rho_d.$$

At $m = 2$, we substitute $a_d = 2/b_d$ and $d = b_d - 2/b_d$ to obtain

$$G_2 - \rho_d = \frac{(b_d^2-2)\big(3b_d^4(b_d-4) + 7b_d^3 + 16b_d^2 + 6(b_d-2)\big)}{2b_d^2(b_d-3)(b_d+1)(3b_d-2)} > 0.$$

For $m \geq 3$, we substitute the definition of $G_m$ into (3.8) and use $a_d b_d = 2$ to obtain

$$\begin{aligned}
&\frac{m(m+1)\big((m+b_d-1)G_m - (m+d)G_{m-1}\big)}{2d\binom{m+d}{d}} \\
&\quad = \frac{(1-2a_d)m^2 + a_d^2(m-1) + a_d}{(m-a_d)(m+1-a_d)} \\
&\qquad + \frac{(m-3)\big((4-a_d)(m+3) + (d-6)(d+1) + a_d(d+2)\big)}{(d+1)(m-2)(m-1)} \\
&\qquad + \frac{(d-9)(d-2) + a_d(d-3) + 2}{(d+1)(m-2)(m-1)} > 0.
\end{aligned} \tag{3.9}$$

The three terms on the right are nonnegative, and the first is positive, since $d \geq 9$, $m \geq 3$, and $0 < a_d < 2/d < 1/2$. Finally, we use

$$\frac{m+d}{m+b_d-1} > 1$$

and (3.9) to obtain, by induction on $m$, $G_m > G_{m-1} > 0$ for every $m \geq 3$. Thus $G_m \geq G_2 > \rho_d$. □

## 3.3 A lower bound for $S_{m,d}(z)$

The following lower bound verifies condition (ii) of Proposition 3.2. Its value at $z = 1$ also supplies the estimate needed in the recurrence argument for $E_{m,d} > 0$ in Subsection 3.4.

**Lemma 3.6.** *For integers $d \geq 10$ and $m \geq 2$, we have*

$$S_{m,d}(z) > \frac{d+3}{b_d+1}\frac{3(d-5)(d-3)(m-1)}{d(d+3+(m-2)z)^4}, \qquad 0 \leq z \leq 1.$$

We first express $S_{m,d}(z)$ in terms of an auxiliary integral. For integers $d \geq 5$ and $m \geq 1$, define

$$I_{m,d}(z) = \int_0^1 t^d(1-t)(1-zt(1-t))^{m-1}\,dt.$$

By expanding $I_{m,d}(z)$ and evaluating the beta integrals, we obtain

$$I_{m,d}(z) = \sum_{j=0}^{m-1}(-1)^j\binom{m-1}{j}\frac{(d+j)!(j+1)!}{(d+2j+2)!}z^j. \tag{3.10}$$

For $m \geq 2$, we use $\vartheta_z = z\,d/dz$ with the same coefficientwise convention as for $\vartheta_x$ and set

$$\Psi(z) = \left(\frac{d+3}{2} + \vartheta_z\right)(3+\vartheta_z)(-I'_{m,d}(z)).$$

Using (3.10) and (3.5), together with the factorization $j(d+j)-2 = (j-a_d)(j+b_d)$, we obtain

$$S_{m,d}(z) = \frac{1}{d}\left(\Psi(z) + (2-a_d)\int_0^1 t^{b_d}\Psi(tz)\,dt\right). \tag{3.11}$$

We use the following expansion and coefficient bound.

**Lemma 3.7.** *For each integer $d \geq 10$, there are real coefficients $\zeta_{d,k}$, $k \geq 0$, such that for every integer $m \geq 2$,*

$$\Psi(z) = (m-1)\sum_{k=0}^{m-2}\binom{m-2}{k}\left(\frac{z}{4}\right)^k\left(1-\frac{z}{4}\right)^{m-2-k}\zeta_{d,k}. \tag{3.12}$$

*Moreover,*

$$\zeta_{d,k} > \frac{3(d-5)(d-3)}{(4k+d+3)^4}, \qquad k \geq 0.$$

*Proof.* We differentiate $I_{m,d}(z)$ and apply the binomial theorem to obtain

$$-\frac{I'_{m,d}(z)}{m-1} = \sum_{k=0}^{m-2}\binom{m-2}{k}\left(\frac{z}{4}\right)^k\left(1-\frac{z}{4}\right)^{m-2-k}v_k,$$

where, for every integer $k \geq 0$, we set

$$\begin{aligned} v_k &= \int_0^1 t^{d+1}(1-t)^2(2t-1)^{2k}\,dt \\ &= 2^{-d-3}\int_0^1 s^{2k}(1-s^2)^2\sum_{h=0}^{n}\binom{d-1}{2h}s^{2h}\,ds. \end{aligned}$$

We obtain the second equality by pairing $t$ with $1-t$ and setting $s = 2t-1$. For real $\alpha$, we apply $\alpha + \vartheta_z$ to replace $v_k$ by $(k+\alpha)v_k - kv_{k-1}$ and omit the second term at $k=0$. We then apply the two operators that define $\Psi(z)$ to obtain (3.12), with coefficients independent of $m$. We evaluate the beta integrals to obtain

$$\zeta_{d,0} = \frac{3}{(d+2)(d+4)}, \qquad \zeta_{d,1} = \zeta_{d,0}\frac{d^2-7d-14}{(d+3)(d+6)}.$$

For $k=0$, we use

$$\zeta_{d,0} > \frac{3}{(d+3)^2} > \frac{3(d-5)(d-3)}{(d+3)^4}.$$

For $k=1$, we combine the identity

$$\begin{aligned} &(d^2-7d-14)(d+9)^2 - (d-5)(d-3)(d+2)(d+3) \\ &\quad = 2(d-10)(7d^2+50d+77) + 316 > 0 \end{aligned}$$

with the arithmetic–geometric mean inequality to obtain

$$\zeta_{d,1} > \frac{3(d-5)(d-3)}{(d+4)(d+6)(d+9)^2} > \frac{3(d-5)(d-3)}{(d+7)^4}.$$

For $k \geq 2$ and $0 \leq h \leq n$, we substitute $t = s^2$ and start with $w(t) = t^{h-1/2}(1-t)^2$. With $\alpha = 3$ and then $\alpha = (d+3)/2$, we integrate by parts using

$$\int_0^1 \bigl((k+\alpha)t^k - kt^{k-1}\bigr)w(t)\,dt = \int_0^1 t^k\,(\alpha w(t) + ((1-t)w(t))')\,dt.$$

We apply both transformations to the weight to obtain

$$\frac{2h-1}{4}t^{h-5/2}(1-t)^3\bigl((2h-3)+(d-2h-2)t\bigr).$$

All boundary terms vanish for $k \geq 2$. At zero, their smallest exponent is $k+h-3/2 \geq 1/2$. At one, they contain $(1-t)^3$ or $(1-t)^4$. The final integrand is at worst of order $t^{-1/2}$ near zero, so the improper integrals converge. Set

$$\Omega_d(t) = \frac{1}{d}\sum_{j=0}^{\lfloor (d+1)/2 \rfloor} (2j-3)(2j-1)(d-2j)\binom{d+1}{2j}t^j.$$

The same formula defines $\Omega_d(t)$ for every integer $d \geq 5$. After summing over $h$ and collecting equal powers of $t$, we obtain

$$\zeta_{d,k} = 2^{-d-6}\int_0^1 t^{k-5/2}(1-t)^3\Omega_d(t)\,dt, \qquad k \geq 2. \tag{3.13}$$

Using $\vartheta_x x^k = kx^k$, we evaluate $\Omega_d(1)$ as

$$\begin{aligned}\Omega_d(1) &= \frac{(\vartheta_x-3)(\vartheta_x-1)(d-\vartheta_x)}{2d}\bigl((1+x)^{d+1}+(1-x)^{d+1}\bigr)\Big|_{x=1}\\ &= (d-5)(d-3)2^{d-3}.\end{aligned}$$

For $1 \leq j \leq \lfloor (d-1)/2 \rfloor$, we use

$$\frac{\binom{d-1}{2j}}{\binom{d+1}{2j}} = \frac{(d+1-2j)(d-2j)}{d(d+1)}, \qquad \frac{\binom{d-1}{2j-2}}{\binom{d+1}{2j}} = \frac{2j(2j-1)}{d(d+1)}$$

to compare the coefficients of $t^j$ in the following recurrence. We check the coefficients for $j = 0$ and $j = \lfloor (d+1)/2 \rfloor$ directly and obtain

$$4(d-3)t\Omega_d'(t) = (d-2)(d+1)\bigl(\Omega_d(t) - (1-t)\Omega_{d-2}(t)\bigr), \qquad d \geq 7. \tag{3.14}$$

For $d \geq 7$, put

$$\kappa_d = \frac{(d-2)(d+1)}{4(d-3)} \leq \frac{d+3}{4}.$$

Using (3.14), we obtain

$$\frac{d}{dt}\bigl(t^{-\kappa_d}\Omega_d(t)\bigr) = -\kappa_d t^{-\kappa_d-1}(1-t)\Omega_{d-2}(t).$$

The initial polynomials $\Omega_5(t) = 3(1-t)^3$ and $\Omega_6(t) = 35(t-1/5)^2 + 8/5$ are positive on $(0,1)$. If $\Omega_{d-2}(t) > 0$ on this interval, then the preceding derivative is negative. Since $\Omega_d(1) > 0$ for $d \geq 7$, we proceed by induction separately for even and odd $d$ to obtain

$$\Omega_d(t) > \Omega_d(1)t^{\kappa_d} \geq \Omega_d(1)t^{(d+3)/4} > 0, \qquad 0 < t < 1.$$

After substituting this bound into (3.13), we evaluate the beta integral and apply the arithmetic–geometric mean inequality to obtain

$$\zeta_{d,k} > \frac{3(d-5)(d-3)}{256(k+(d-3)/4)_4} > \frac{3(d-5)(d-3)}{(4k+d+3)^4}, \qquad k \geq 2.$$

Combining this bound with the cases $k = 0, 1$, we obtain the asserted inequality. □

*Proof of Lemma 3.6.* We interpret the weights in (3.12) as the probabilities of a binomial distribution with parameters $m-2$ and $z/4$, whose mean is $(m-2)z/4$. Using Lemma 3.7 and Jensen's inequality for the convex function $u \mapsto (4u+d+3)^{-4}$ on $[0,\infty)$, we obtain

$$\Psi(z) > \frac{3(d-5)(d-3)(m-1)}{(d+3+(m-2)z)^4}.$$

The inequality is also strict when $m=2$ or $z=0$, since the bounds on the coefficients are strict. Since $tz \le z$ for $t, z \in [0,1]$, we substitute this bound into (3.11) and use

$$1 + \frac{2-a_d}{b_d+1} = \frac{d+3}{b_d+1}$$

to conclude the proof. □

## 3.4 Positivity of $E_{m,d}$

The following proposition verifies condition (iii) of Proposition 3.2.

**Proposition 3.8.** *For integers $d \ge 17$ and $m \ge 2$, we have $E_{m,d} > 0$.*

We use a direct estimate for small coranks. For larger coranks, we work backwards through a recurrence for $E_{m,d}$.

Differentiating $g(t)$ at zero and using (3.7), we obtain the following expression for $E_{m,d}$. We also use it to define $E_{1,d}$.

$$E_{m,d} = \frac{1}{md\binom{m+d}{d}} \left( \phi_d\big(R_{m,d}(b_d) - d\bar{\lambda}_{m,d}\big) - \frac{\eta_d}{b_d+1} - \frac{1}{2}\sum_{j=1}^{m-1} c^{\mathrm{R}}_{m,d}(j)(d+1+2j) \right). \tag{3.15}$$

For $2 \le m \le d$, set

$$B_{m,d} = \phi_d\left( \frac{a_d+1}{m(d+1)} - \frac{a_d+2}{(m+b_d)(m+b_d+1)} \right) + \frac{\eta_d}{md\binom{m+d}{d}}. \tag{3.16}$$

For $m \ge d+1$, set

$$B_{m,d} = \phi_d\left( \frac{a_d+2}{m(m+1)} - \frac{a_d+2}{(m+b_d)(m+b_d+1)} \right) + \frac{\eta_d}{md\binom{m+d}{d}}. \tag{3.17}$$

The next lemma reduces positivity for large coranks to a comparison between $S_{m,d}(1)$ and $B_{m,d}$.

**Lemma 3.9.** *Fix an integer $d \ge 5$ and an integer $m_0 \ge 2$. If $S_{m,d}(1) > B_{m,d}$ for every $m \ge m_0$, then $E_{m,d} > 0$ for every $m \ge m_0 - 1$.*

*Proof.* We first derive a recurrence for $E_{m,d}$. We substitute $u = b_d = d + a_d$ into (3.8) to obtain

$$(m-1-a_d)R_{m,d}(b_d) = (m+d)R_{m-1,d}(b_d) - \frac{md(a_d+2)\binom{m+d}{d}}{(m+b_d)(m+b_d+1)}.$$

We set $c^{\mathrm{R}}_{m-1,d}(m-1) = 0$ and use (3.5) to obtain

$$(m+d)c^{\mathrm{R}}_{m-1,d}(j) = (m-1-j)c^{\mathrm{R}}_{m,d}(j), \qquad 1 \le j \le m-1.$$

The terms involving $\bar{\lambda}_{m,d}$ satisfy

$$\frac{1}{\max(m-1,d)+1} - \frac{m-1-a_d}{m(\max(m,d)+1)} = \begin{cases} (a_d+1)/(m(d+1)), & m \le d, \\ (a_d+2)/(m(m+1)), & m \ge d+1. \end{cases}$$

The second branch also holds at $m = d+1$, since $\max(m-1,d)+1 = m$. We divide the first two identities by $md\binom{m+d}{d}$ and substitute them, together with the preceding identity, into (3.15). Using

$$\binom{m+d-1}{d} = \frac{m}{m+d}\binom{m+d}{d},$$

we obtain

$$(m-1-a_d)E_{m,d} = (m-1)E_{m-1,d} - S_{m,d}(1) + B_{m,d}, \qquad m \geq 2. \tag{3.18}$$

We next obtain a lower bound for $E_{m,d}$ by retaining only the possibly negative terms in (3.15). We substitute $u = b_d$ into the partial fraction expansion of

$$\frac{R_{m,d}(u)}{md\binom{m+d}{d}}.$$

By the residue formulas in the proof of Lemma 3.1, the poles $-1, \ldots, -m$ contribute positive terms, whereas the pole $-m-1$ contributes

$$-\frac{d+m-1}{(d+2m)(b_d+m+1)}.$$

For $0 \leq j \leq m-1$, the coefficient of $(d+j-u)^{-1}$ equals $-(j+2)$ times the coefficient of $z^j$ in $I_{m,d}(z)$. In (3.15), we multiply these contributions from the positive poles by $\phi_d$ and combine them with the alternating sum, using (3.5), (3.10), and the identity

$$\begin{aligned}(j+2)&\left(\frac{j(d+j+2)(d+1+2j)}{2d(j(d+j)-2)} - \frac{\phi_d}{j-a_d}\right)\\ &= \frac{1}{d}\left(j(j-1) + \frac{d+11}{2}j + 8 - d\phi_d + \frac{(d-3+d\phi_d)(b_d-2)}{j+b_d}\right).\end{aligned}$$

This identity also holds at $j=0$. We sum these terms to express their contribution to $E_{m,d}$ as

$$\begin{aligned}&\frac{I''_{m,d}(1)}{d} + \frac{d+11}{2d}I'_{m,d}(1) + \left(\frac{8}{d} - \phi_d\right)I_{m,d}(1)\\ &\qquad + \left(1 - \frac{3}{d} + \phi_d\right)(b_d-2)\int_0^1 z^{b_d-1}I_{m,d}(z)\,dz.\end{aligned}$$

The integral defining $I_{m,d}(z)$ gives $I_{m,d}(z) \geq 0$ on $[0,1]$ and $I''_{m,d}(1) \geq 0$. Moreover, $b_d > 2$. The definition of $\phi_d$, together with $a_d < 2/d$ and $a_d + b_d > d$, gives

$$\frac{1}{d} < \phi_d < \frac{(d+1)(d+2)}{d^3}. \tag{3.19}$$

In particular, $d\phi_d < 3$ for $d \geq 5$. In the displayed sum of four terms, only the term involving $I'_{m,d}(1)$ can be negative. For $0 \leq t \leq 1$, we apply the binomial theorem to obtain

$$\binom{m}{2}t^2(1-t)^2(1-t(1-t))^{m-2} \leq 1.$$

We multiply by $2t^{d-1}/m$ and integrate to obtain

$$0 \leq -I'_{m,d}(1) = (m-1)\int_0^1 t^{d+1}(1-t)^2(1-t(1-t))^{m-2}\,dt \leq \frac{2}{md}.$$

We discard the other three terms in that sum and the positive terms from the negative poles. Since $(d+m-1)/(d+2m) < 1$, $b_d + m + 1 > m$, and $\binom{m+d}{d} \geq 1$, we obtain

$$E_{m,d} \geq -\frac{1}{m}\left(\frac{d+11}{d^2} + 2\phi_d + \frac{\eta_d}{d(b_d+1)}\right). \tag{3.20}$$

Fix integers $k \geq m_0 - 1$ and $L > k$. By hypothesis, $S_{m,d}(1) > B_{m,d}$ for $m \geq k+1$. We iterate (3.18) and retain the term with $m = k+1$ to obtain

$$E_{k,d} \geq \left(\prod_{j=k}^{L-1}\left(1 - \frac{a_d}{j}\right)\right)E_{L,d} + \frac{S_{k+1,d}(1) - B_{k+1,d}}{k}.$$

Since $0 < a_d < 1$ and $k \geq 1$, the product lies in $(0,1]$. Using (3.20) and letting $L \to \infty$, we conclude that

$$E_{k,d} \geq \frac{S_{k+1,d}(1) - B_{k+1,d}}{k} > 0.$$

□

*Proof of Proposition 3.8.* We prove $E_{m,d} > 0$ directly for $2 \le m \le (d-2)/3$ and then verify the hypothesis of Lemma 3.9 for $m \ge \lceil (d+1)/3 \rceil$. The lemma then covers $m \ge \lfloor d/3 \rfloor$, and the direct estimate covers all remaining integers $m \ge 2$. We use (3.19) throughout the proof and also need $\eta_d < d$. In the defining formula for $\eta_d$, each numerator factor is less than $d+5 \le 22d/17$, and the denominator exceeds $3d^3$. Hence

$$\eta_d < \frac{1}{3}\left(\frac{22}{17}\right)^4 d < d.$$

First suppose $2 \le m \le (d-2)/3$. We use (3.15) to bound $E_{m,d}$ from below by estimating the term containing $R_{m,d}(b_d)$ and the alternating sum separately. We first prove by induction that

$$\frac{R_{m,d}(b_d)}{md\binom{m+d}{d}} > \frac{1}{d+2}, \qquad m \ge 2.$$

At $m = 2$, we evaluate the left-hand side as

$$\frac{b_d - 1}{(b_d - 2)(b_d + 3)} > \frac{1}{d+2}.$$

For $m \ge 3$, we use the same normalization of (3.8) at $u = b_d$ as in the proof of Lemma 3.9. Applying the induction hypothesis, we reduce the desired bound to

$$\frac{m-1}{d+2} - \frac{a_d + 2}{(m + b_d)(m + b_d + 1)} > \frac{m - 1 - a_d}{d+2}.$$

We use $a_d(b_d + 1) = a_d + 2$ and $(m + b_d)(m + b_d + 1) > (b_d + 1)(d + 2)$ to verify this inequality. Since $m - 1 - a_d > 0$, we complete the induction. Also,

$$\frac{\eta_d}{b_d + 1} < 1, \qquad m\binom{m+d}{d} \ge (d+1)(d+2).$$

We retain the negative prefactor and write the last term in (3.15) as an alternating sum. For $m = 2$, the sum consists of its $j = 1$ term. For $m \ge 3$, we use (3.5) to compare the absolute values of consecutive summands. One positive factor decreases with $j \ge 1$, since

$$\frac{d}{du}\left(\frac{u(d+u+2)}{d(u(d+u)-2)}\right) = -\frac{2(u^2+2u+d+2)}{d(u^2+du-2)^2} < 0, \qquad u \ge 1.$$

For $1 \le j \le m-2$, we divide the product of the remaining factors at $j+1$ by its value at $j$ to obtain

$$(m-1-j)\frac{j+3}{j+1}\frac{d+j+1}{(d+2j+4)(d+2j+1)} < \frac{2(m-2)}{d+6} < 1.$$

We therefore bound the alternating sum from below by its $j = 1$ term. We also use $\phi_d > 1/d$ and

$$\frac{1}{d+2} - \frac{1}{m(d+1)} > 0.$$

We substitute these estimates into (3.15) and use $\max(m, d) = d$ to obtain

$$E_{m,d} > \frac{1}{(d+1)(d+2)} - \frac{1}{d(d+1)m} - \frac{3(d+3)(m-1)}{d(d-1)(d+2)(d+4)}.$$

The right-hand side is concave as a function of $m$. At $m = 2$ and $m = (d-2)/3$, respectively, it has the same sign as

$$d^3 - 5d^2 - 34d - 10 \qquad \text{and} \qquad d^2 - 11d - 6.$$

Both polynomials are increasing for $d \ge 17$ and positive at $d = 17$. Thus the lower bound is positive throughout the interval, which proves $E_{m,d} > 0$ in the stated range.

To apply Lemma 3.9, now suppose $m \ge \lceil (d+1)/3 \rceil$. Using $b_d < d + 2/d$, we apply Lemma 3.6 to obtain

$$\frac{(m+d+1)^4 S_{m,d}(1)}{m-1} > \frac{3(d-5)(1-9/d^2)}{1+1/d+2/d^2} \ge \frac{360}{11} > \frac{129}{4},$$

since each of the positive factors $d-5$, $1-9/d^2$, and $(1+1/d+2/d^2)^{-1}$ increases with $d \geq 17$. After multiplying the two terms of $B_{m,d}$ by

$$\frac{(m+d+1)^4}{m-1},$$

we bound the term containing $\eta_d$ by $1/4$ and the term containing $\phi_d$ by 32.

For the term containing $\eta_d$, put $s = m+d \geq 23$. Since $m \geq 6$, $\eta_d < d$, and $\binom{s}{d} \geq \binom{s}{6}$, the product formula for $\binom{s}{6}$ gives

$$\frac{\eta_d (s+1)^4}{m(m-1)d\binom{s}{d}} < \frac{24(s+1)^4}{(s-5)^6} \leq \frac{24}{18^2}\left(\frac{4}{3}\right)^4 < \frac{1}{4}.$$

For the term containing $\phi_d$, we use (3.19) and $a_d < 2/d$. First suppose $(d+1)/3 \leq m \leq d$, and put

$$q = \frac{(m+d+1)^2}{m(d+1)}.$$

Then

$$4 \leq q \leq \frac{16}{3}, \qquad \frac{m}{m-1} \leq \frac{6}{5}.$$

Since $a_d < 1/2$ gives $(m+b_d)(m+b_d+1) < (m+d+1)^2$, the $\phi_d$ contribution in (3.16) satisfies

$$\begin{aligned}
&\frac{(m+d+1)^4 \phi_d}{m-1}\left(\frac{a_d+1}{m(d+1)} - \frac{a_d+2}{(m+b_d)(m+b_d+1)}\right) \\
&< \phi_d(d+1)\frac{m}{m-1}\, q\big((a_d+1)q-(a_d+2)\big) \\
&< \frac{32}{15}\left(1+\frac{1}{d}\right)^2\left(1+\frac{2}{d}\right)\left(10+\frac{26}{d}\right) \\
&\leq \frac{32}{15}\left(\frac{18}{17}\right)^2 \frac{19}{17}\frac{196}{17} < 32.
\end{aligned}$$

Here $q((a_d+1)q-(a_d+2))$ is positive and increasing in both $q \geq 4$ and $a_d > 0$.

For $m \geq d+1$, we multiply the $\phi_d$ term in (3.17) by the same factor and use the same denominator comparison to bound it strictly by

$$\begin{aligned}
&\phi_d(a_d+2)\frac{(m+d+1)^2}{m(m+1)}\,\frac{(2d+1)m+(d+1)^2}{m-1} \\
&\leq \frac{4\phi_d(a_d+2)(d+1)^2(3d+2)}{d(d+2)} \\
&< 8\left(1+\frac{1}{d}\right)^4\left(3+\frac{2}{d}\right) \leq 8\left(\frac{18}{17}\right)^4 \frac{53}{17} < 32.
\end{aligned}$$

The two factors involving $m$ decrease with $m$, so we evaluate them at $m = d+1$, then apply (3.19) and $a_d < 2/d$. The remaining factors decrease with $d$, so we evaluate them at $d = 17$ to obtain the last bound. In both cases, we therefore obtain

$$B_{m,d} < \frac{129(m-1)}{4(m+d+1)^4} < S_{m,d}(1).$$

Lemma 3.9 now gives $E_{m,d} > 0$ for $m \geq \lfloor d/3 \rfloor$ and completes the proof. □

*Proof of Proposition 2.6 for $d \geq 17$.* We apply Proposition 3.5, Lemma 3.6, and Proposition 3.8 to verify the conditions (i), (ii), and (iii) of Proposition 3.2, respectively, and obtain (2.9). □

# 4 Exact certificates for $5 \leq d \leq 16$

We first prove Proposition 2.6 for $5 \leq d \leq 16$. We check the signs of finitely many polynomials to treat every $m \geq 2$. We then prove Theorems 1.2 and 1.1.

For integers $d \geq 5$ and $m \geq 2$, set

$$\widehat{\lambda}_{m,d} = \frac{(m+2)_{d-1}}{d!} - \frac{120(d-m)(m+5)_{d-4}}{(d+1)d!}. \tag{4.1}$$

We define

$$H_{m,d}(x) = \frac{P_{m,d}(x) - \widehat{\lambda}_{m,d}\Delta_d(x) - P_{2,d-2}(x)}{x}.$$

For fixed $d$, the quantity $\widehat{\lambda}_{m,d}$ is a polynomial in $m$. We can also write

$$\widehat{\lambda}_{m,d} = \binom{m+d}{d}\left(\frac{1}{m+1} + \frac{120}{(m+2)_3}\left(\frac{1}{d+1} - \frac{1}{m+1}\right)\right).$$

Since

$$0 < \frac{120}{(m+2)_3} \leq 1,$$

the parenthesized factor is a convex combination of $1/(m+1)$ and $1/(d+1)$. Hence $\widehat{\lambda}_{m,d} \geq \bar{\lambda}_{m,d}$, with equality if and only if $m = 2$ or $m = d$. Subtracting $P_{2,d-2}(x)$ cancels the constant term without changing values at its zeros. Thus $H_{m,d}(x)$ is a polynomial of degree at most $n-1$. For every zero $r$ of $P_{2,d-2}(x)$, equation (2.7) gives

$$\frac{F_{m,d}(r)}{rP'_{2,d-2}(r)} \geq \frac{H_{m,d}(r)}{P'_{2,d-2}(r)}. \tag{4.2}$$

Thus it suffices to determine the signs of the remainder of $H_{m,d}(x)$ modulo $P_{2,d-2}(x)$.

**Lemma 4.1.** *For each integer $d \geq 5$, the remainder of $H_{m,d}(x)$ modulo $P_{2,d-2}(x)$ admits a polynomial extension in $m$ of degree at most $d-2$.*

*Proof.* Fix $1 \leq k \leq n$. We rewrite the binomial coefficients in (2.1) with lower arguments $d-2k-1$ and $k-1$ to express the summand as a polynomial in $h$ of degree $d-k-2$, with leading coefficient

$$\frac{1}{(d-2k-1)!(k-1)!}.$$

Summing over $0 \leq h < m$ raises the degree by one and divides the leading coefficient by $d-k-1$. After multiplying by the prefactor in (2.1), we find that $[x^k]P_{m,d}(x)$ is a polynomial in $m$ of degree $d-1$, with leading coefficient

$$\frac{\binom{d-k-2}{k-1}}{k!(d-k)!}.$$

Comparing with (2.4), we find that the coefficient of $m^{d-1}$ in the polynomial

$$\frac{P_{m,d}(x) - 1}{x}$$

is

$$\frac{P_{1,d-2}(x)}{(d-1)!}.$$

The second term in (4.1) has degree $d-3$. We therefore compute the coefficient of $m^{d-1}$ in $H_{m,d}(x)$ as

$$\frac{dP_{1,d-2}(x) - \Delta_d(x)/x}{d!} = \frac{P_{2,d-2}(x)}{d!}.$$

We reduce modulo $P_{2,d-2}(x)$ to eliminate this term and obtain the claimed degree bound. □

We expand this remainder in powers of $m-2$. Lemma 4.1 shows that only powers with exponents from 0 to $d-2$ occur, and each power is nonnegative for $m \geq 2$. In view of (4.2), it therefore suffices to show that all $d-1$ polynomials in this expansion have the same sign as $P'_{2,d-2}(r)$ at each zero $r$ of $P_{2,d-2}(x)$. We check these signs on rational intervals containing the zeros. For $5 \leq d \leq 16$ and $0 \leq j \leq d-2$, consider the coefficient of $(m-2)^j$ in the remainder of $H_{m,d}(x)$ modulo $P_{2,d-2}(x)$. Let $p_{d,j}(x)$ be the primitive integer polynomial that is a positive rational multiple of this coefficient. All these polynomials are nonzero and have degree less than $n-1$. By Lemma 4.1, we recover the remainder by Lagrange interpolation from its values at $m = 2, \ldots, d$. Equations (2.1) and (4.1) therefore determine the polynomials $p_{d,j}(x)$ by exact rational arithmetic.

*Proof of Proposition 2.6 for* $5 \le d \le 16$. We first locate the zeros of $P_{2,d-2}(x)$. Let $(A_i, q_i)$, $1 \le i \le n-1$, denote the pairs in Table 1, in the displayed order. They define disjoint intervals

$$\left(\frac{A_i}{q_i}, \frac{A_i+1}{q_i}\right), \qquad 1 \le i \le n-1.$$

We substitute the endpoints into $P_{2,d-2}(x)$ to obtain

$$(-1)^{n-1-i}P_{2,d-2}(A_i/q_i) < 0 < (-1)^{n-1-i}P_{2,d-2}((A_i+1)/q_i).$$

Consequently the $i$th interval contains the $i$th zero $r_i$ of $P_{2,d-2}(x)$, and $(-1)^{n-1-i}P'_{2,d-2}(r_i) > 0$.

To check the signs of $p_{d,j}(x)$ on these intervals, we form the integer polynomials in $y$

$$(-1)^{n-1-i}q_i^{n-2}(1+y)^{n-2}p_{d,j}\left(\frac{A_i+(A_i+1)y}{q_i(1+y)}\right). \tag{4.3}$$

The substitution in (4.3) maps $y > 0$ onto the $i$th interval. Every coefficient of this transformed polynomial is a positive integer, so $(-1)^{n-1-i}p_{d,j}(x) > 0$ throughout that interval. The last column of Table 1 gives the minimum of all coefficients of these transformed polynomials over $0 \le j \le d-2$ and $1 \le i \le n-1$. For example, when $d = 7$ and $j = 0$, we have $p_{7,0}(x) = 163x + 38$. For the two intervals $(-2,-1)$ and $(-1/16, 0)$, (4.3) gives $288 + 125y$ and $445 + 608y$, respectively. The supplementary material gives the explicit expansion of (4.3) and includes a program for exact computation and its output, with all primitive polynomials, transformed coefficients, endpoint values, and minima in Table 1.

Table 1. Rational intervals and minimum positive coefficients in (4.3). A pair $(A, q)$ denotes $(A/q, (A+1)/q)$.

| $d$ | $(A_1, q_1), \ldots, (A_{n-1}, q_{n-1})$ | Least coefficient |
|---|---|---|
| 5 | $(-1,1)$ | 1 |
| 6 | $(-1,1)$ | 1 |
| 7 | $(-2,1),\ (-1,16)$ | 24 |
| 8 | $(-4,8),\ (-1,32)$ | 117 |
| 9 | $(-4,1),\ (-3,8),\ (-1,64)$ | 170 |
| 10 | $(-2,1),\ (-2,8),\ (-1,64)$ | 4078 |
| 11 | $(-6,1),\ (-2,2),\ (-5,32),\ (-1,128)$ | 672 |
| 12 | $(-4,2),\ (-6,16),\ (-4,32),\ (-1,128)$ | 1723155 |
| 13 | $(-10,1),\ (-4,4),\ (-5,16),\ (-3,32),\ (-1,256)$ | 447905 |
| 14 | $(-3,1),\ (-2,2),\ (-4,16),\ (-5,64),\ (-1,256)$ | 329123 |
| 15 | $(-14,1),\ (-2,1),\ (-4,8),\ (-6,32),\ (-4,64),\ (-3,1024)$ | 16890 |
| 16 | $(-4,1),\ (-4,4),\ (-6,16),\ (-5,32),\ (-7,128),\ (-1,512)$ | 54329359 |

We evaluate the remainder at $r_i$ to recover $H_{m,d}(r_i)$. For $0 \le j \le d-2$, its coefficient of $(m-2)^j$ is a positive multiple of $p_{d,j}(r_i)$. Thus every coefficient of $(-1)^{n-1-i}H_{m,d}(r_i)$ is positive in these degrees, and all higher coefficients vanish. In particular, the coefficient with $j = 0$ is positive. Consequently,

$$\frac{H_{m,d}(r_i)}{P'_{2,d-2}(r_i)} > 0 \qquad \text{for every } m \ge 2.$$

We now apply (4.2) to conclude the proof. □

We can now prove the main theorems.

*Proof of Theorem 1.2.* Let $\lambda \le \bar{\lambda}_{m,d}$. We write

$$P_{m,d}(x) - \lambda\Delta_d(x) = F_{m,d}(x) + (\bar{\lambda}_{m,d} - \lambda)\Delta_d(x).$$

Since $\bar{\lambda}_{m,d} - \lambda \ge 0$, Proposition 2.5 and (2.8) show that this polynomial has degree $n$ and positive leading coefficient. Its constant term is 1. Proposition 2.6 and (2.7) show that the rational function

$$\frac{P_{m,d}(x) - \lambda\Delta_d(x)}{xP_{2,d-2}(x)}$$

has positive residues at every zero of $P_{2,d-2}(x)$. Its residue at zero is 1 because $P_{2,d-2}(0) = 1$. We apply Lemma 2.1(ii) to obtain

$$P_{m,d}(x) - \lambda\Delta_d(x) \prec xP_{2,d-2}(x).$$

Since the largest zero of $xP_{2,d-2}(x)$ is zero, we conclude that $P_{m,d}(x) - \lambda\Delta_d(x)$ has only simple negative zeros and that $P_{2,d-2}(x)$ strictly interlaces this polynomial. □

*Proof of Theorem 1.1.* Let $M$ be a sparse paving matroid of rank $d$ and corank $m$, and let $\lambda$ be its number of circuit-hyperplanes. Suppose that $P_M(x)$ is nonconstant. Then $M$ is loopless and $P_M(0) = 1$. For $d \leq 4$, the degree bound and nonnegative coefficients [8, Theorems 2.5 and 5.6] imply that $P_M(x)$ is linear with positive coefficients. Its unique zero is simple and negative.

We may therefore assume $d \geq 5$. Corank zero gives the free matroid and the constant polynomial 1. When $m = 1$, two distinct $d$-subsets of the ground set meet in $d-1$ elements, whereas distinct circuit-hyperplanes meet in at most $d-2$ elements. Hence $\lambda$ is either zero or one, and the relaxation formula gives $P_{1,d}(x)$ or $P_{1,d-1}(x)$. We apply Lemma 2.4 with parameter $d+2$ and $d+1$, respectively, to prove the assertion in either case.

For $m \geq 2$, the bound on $\lambda$ and the relaxation formula give $P_M(x) = P_{m,d}(x) - \lambda\Delta_d(x)$ with $0 \leq \lambda \leq \bar{\lambda}_{m,d}$. We therefore apply Theorem 1.2 to complete the proof. □

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## Declaration of Generative AI

During the preparation of this manuscript, the author used ChatGPT Pro to explore proof strategies, write programs, and assist with writing. The author takes full responsibility for the content of this manuscript.

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